

The Problem : Sorting

Given: an unsorted array A
(of real number)

want: A to be sorted,
from smallest to largest

Notes: Describing the problem
describes WHAT we need to
solve, but does not prescribe
HOW to do it, or HOW FAST it
should be done.

Note:

We

real

mode

- data

- RAM

20

"

access

array;

Linked L

Note:

We use the
Real ($\mathbb{R}$)-RAM

model of computation

- data we store: real #s
- RAM: if I know a memory address, I can access it in $\Theta(1)$ time.
"constant"

access the i th element
array: $\Theta(1)$
Linked List: $\Theta(i)$

Algo Te
Jeff E
(free

Sorting Algor:

- Quick Sort $\Theta(n^2)$ worst-case
 $\Theta(n \log n)$ expected
- Bubble Sort $\Theta(n^2)$
- pairwise compare
- Selection Sort: $\Theta(n^2)$
- find biggest, then next biggest...
- Merge Sort $\Theta(n \log n)$
- divide + conquer
- recursion: $T(n) = 2T(n/2) + \Theta(n)$
 $\in \Theta(n \log n)$

Algo Text:

Jeff Erickson's Algorithms
(free online)

/"The Algorithms Bible"
CLRS Introduction to Algorithms
(4 authors)

Sets: A, B

a relation: a subset of $A \times B$

$$(A \times B := \{(a, b) \mid a \in A, b \in B\})$$

Notationally: $(a, b) \in R$ "ordered pairs",
 $a R b$ "infix"

Equivalence relation: a relation R such that:

(i) transitivity

$$a R b, b R c \Rightarrow a R c$$

(ii) reflexivity

$$a R a \in R$$

(iii) symmetry

$$a R b \Rightarrow b R a$$

Notationally: $\sim$ $\backslash \text{sim}$

Topo Spaces

- $\mathbb{R}, \mathbb{N}, \mathbb{Z}, \mathbb{Z}_+$
 $\mathbb{N} = \{0, 1, 2, \dots\}$

- $S^n := \{p \in \mathbb{R}^{n+1} \mid \|p\|_2 = 1\}$

- $S^n / \sim$, where $a \sim b$ if they are antipodal
 $\hookrightarrow \mathbb{RP}^n$ projective space! $\cong \{l \mid l \text{ goes through } \vec{0}\}$

- identification diagrams

$$T \cong \text{square with arrows on edges}, T \cong S^1 \times S^1 \subseteq \mathbb{R}^4$$

proj into $\mathbb{R}^3$ 